SMART VIT HACKATHON 2026

AI-Powered Smart Healthcare Platform for Accessible and Preventive Healthcare

- **Problem Statement ID – SVH26006**
- **Problem Statement Title- AI-Driven Public Health Chatbot for disease awareness**
- **Theme- MedTech / BioTech / HealthTech**
- **PS Category- Software**
- **Team ID- svh-10088**
- **Team Name (Registered on portal) : AC-DC**

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AI HealthConnect

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INNOVATION SE AATMANIRBHAR BHARATH

PROPOSED SOLUTION

An AI-powered, multilingual healthcare platform that makes healthcare accessible, secure, and proactive for everyone.

DETAILED EXPLANATION OF THE PROPOSED SOLUTION

Conversational AI on WhatsApp
Users interact via chat or voice in their preferred language for any health query.

AI Scans & Understands
Computer vision & OCR extract data from reports, images & prescriptions.

AI Provides Insights
Generates easy-to-understand summaries, medication explanations & health advice.

Secure Record Storage
All data is stored on blockchain - safe, transparent & under patient control.

Predict & Prevent
Predictive analytics and heatmaps help users stay ahead of potential health risks.

Take Action
Users can connect to hospitals, access schemes, follow diet plans & get reminders.

WhatsApp AI Health Assistant

Chat | Voice | Local Languages

AI Medical & Prescription Scanner
Scans reports, X-rays, MRIs & handwritten prescriptions and converts them into easy summaries.

Emergency Assistance
1-tap SOS, live location sharing & nearby hospitals, pharmacies, ambulances.

Personalized Nutrition Advisor
AI-based diet & nutrition plans based on your health reports and risk profile.

Smart Reminders & Health Broadcasts
Medication, water intake, appointment reminders & real-time health alerts via WhatsApp.

Blockchain Health Records
Patient-owned, tamper-proof storage for medical history, prescriptions & vaccination records.

Outbreak Heatmap & Predictive Analytics
Live disease heatmap & personalized risk prediction based on location & health data.

Mental Health Support Hub
AI-guided support, self-assessment & nearby mental health expert mapping.

HOW IT ADDRESSES THE PROBLEM

Bridges Healthcare Access Gap
Provides healthcare guidance to rural, underserved & non-English speaking communities via WhatsApp.

Simplifies Complex Medical Information
AI turns complicated reports and prescriptions into simple, actionable insights.

Secures & Empowers Patients
Blockchain ensures data privacy, ownership and secure sharing of medical records.

Prevents & Predicts Health Risks
Live outbreak tracking & risk prediction help users take early preventive actions.

Saves Lives in Emergencies
Instant location sharing and hospital discovery improve emergency response time.

Promotes Holistic Well-being
Mental health support, nutrition advice and smart reminders improve overall health.

Connects to Government Support
Users can easily find and apply for relevant health schemes and benefits.

INNOVATION AND UNIQUENESS OF THE SOLUTION

WhatsApp-First Healthcare
Using the most popular app to deliver AI healthcare in local languages.

AI + Computer Vision Integration
Unique combination to scan, read & explain medical reports and prescriptions.

Blockchain for Health Records
True patient ownership, security and transparency of medical data.

Outbreak Heatmap + Personal Risk
Real-time disease tracking with personalized risk prediction.

Complete Health Ecosystem
Combines emergency, mental health, nutrition, reminders & schemes in one platform.

Inclusive & Accessible Design
Voice support, regional languages & guided onboarding for all age groups.

AI HealthConnect brings the power of AI, Trust of Blockchain, and Ease of WhatsApp to make healthcare SMART, SECURE & FOR EVERYONE.

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TECHNICAL APPROACH

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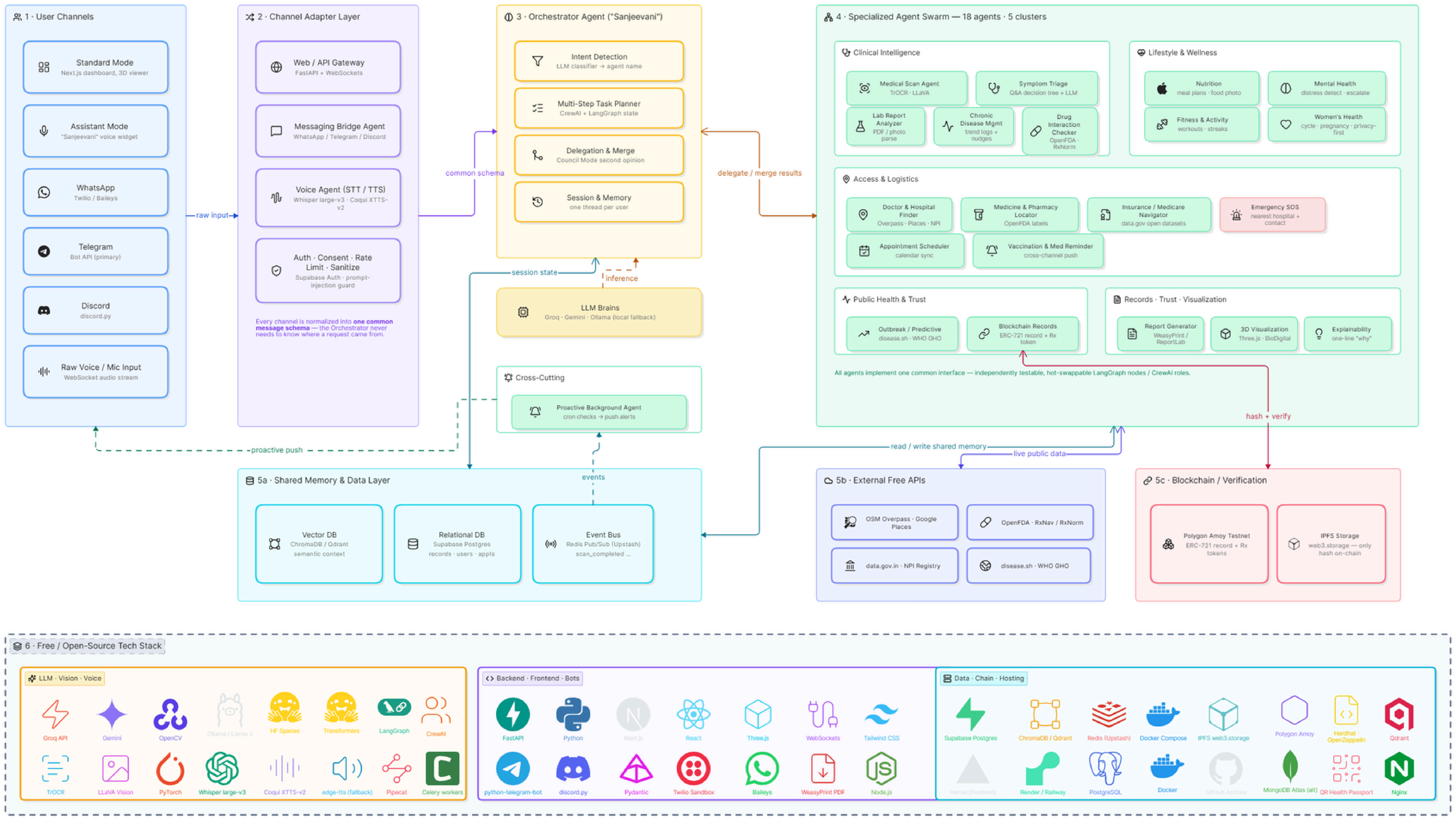

Architecture Diagram

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FEASIBILITY AND VIABILITY

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FEASIBILITY
Why the idea is feasible

Existing AI & OCR Technology
Mature AI models for image recognition, NLP & prediction.

WhatsApp-Based Accessibility
High penetration ensures wide reach & easy adoption.

Cloud & Scalable Infrastructure
Reliable cloud services allow scalability & high availability.

Secure Blockchain Records
Proven blockchain tech for tamper-proof, patient-owned health records.

CHALLENGES & RISKS
What might stop us

Data Privacy
Sensitive health data requires high security and compliance.

AI Accuracy
Incorrect predictions may impact trust and decision-making.

Hospital Integration
Different systems and protocols may slow adoption.

Internet Connectivity
Limited connectivity in rural/remote areas.

OUR STRATEGY
How we overcome them

End-to-End Encryption
Strong encryption, consent management & role-based access control.

Doctor Verification
Verified healthcare professionals ensure reliable responses.

Continuous AI Training
Model retraining with real data & feedback for better accuracy.

Offline & Multilingual Support
Works in low-bandwidth conditions and supports local languages.

PROJECT VIABILITY

TECHNICALLY FEASIBLE
Built on proven technologies.

COST EFFECTIVE
Leverages open-source tools and cloud.

SCALABLE
Designed to grow with users & data.

HIGH SOCIAL IMPACT
Improves healthcare access, outcomes & equity.

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IMPACT AND BENEFITS

Creating a healthier, empowered and accessible society for all.

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IMPACT ON TARGET AUDIENCE

Accessible Healthcare for All
Brings quality healthcare to rural, underserved & remote communities.

Faster & Smarter Decisions
AI insights and real-time support enable quick diagnosis and informed choices.

Empowered Individuals
Easy-to-understand information, reminders and guidance in local languages.

Better Health Outcomes
Early detection, timely treatment and continuous care lead to healthier lives.

SOCIAL BENEFITS

Improved Public Health
Predictive analytics & outbreak tracking help control disease spread.

Reduced Health Inequality
Technology + local language support bridges the urban-rural gap.

Mental Well-being Support
Easy access to mental health resources and nearby support networks.

Stronger Communities
Health awareness, education and engagement create healthier communities.

ECONOMIC BENEFITS

Lower Healthcare Costs
Early diagnosis and remote guidance reduce hospital visits and treatment costs.

Higher Productivity
Healthier individuals mean fewer sick days and more productivity.

Savings for Families
Affordable access to healthcare and medication reminders reduce expenses.

Efficient Resource Use
Optimized use of hospitals, medicines and healthcare resources.

ENVIRONMENTAL BENEFITS

Paperless Healthcare
Digital records, e-prescriptions and reports reduce paper usage.

Reduced Carbon Footprint
Tele-consultation & remote support reduce travel and emissions.

Sustainable Operations
Cloud-based and efficient systems minimize resource wastage.

Environmentally Conscious Society
Promotes digital adoption and sustainable healthcare practices.

AI HealthConnect | Transforming Healthcare. Empowering Lives. Building a Better Tomorrow.

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RESEARCH AND REFERENCES

Fang, H. S. A., Tan, T. H., Tan, Y. F. C., & Tan, C. J. M. (2021). Blockchain Personal Health Records: Systematic Review. Journal of Medical Internet Research, 23, e25094. https://doi.org/10.2196/25094 Cited by: 129

Kiania, K., Jameii, S. M., & Rahmani, A. M. (2023). Blockchain-based privacy and security preserving in electronic health: a systematic review. Multimedia Tools and Applications, 82, 28493–28519. https://doi.org/10.1007/s11042-023-14488-w Cited by: 121

Aiken, E. L., McGough, S. F., Majumder, M. S., et al. (2020). Real-time estimation of disease activity in emerging outbreaks using internet search information. PLOS Computational Biology, 16, e1008117. https://doi.org/10.1371/journal.pcbi.1008117 Cited by: 36

Dinesh, A. S., Mathur, V., Ansil, B. R., et al. (2020). Health Heatmap of India: An Open Data Platform. Journal of the Indian Institute of Science, 100, 701–716. https://doi.org/10.1007/s41745-020-00196-z Cited by: 7